CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
FACULTY OF TECHNOLOGY AND ENGINEERING (FTE)
SUBJECT: WEB DEVELOPMENT FRAMEWORKS (ITUE203)
SEMESTER: 3RD, 2026-27 (ODD)

1. Practical Summary

No.	Practical Title	Core Technology/Concept	CO Mapping	PO Mapping	Hours
1	Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup	Requirements, wireframe, GitHub	CO1	PO1(3), PO2(2), PO3(3), PO4(2), PO5(3), PO9(1), PO10(3)	04
2	Semantic HTML5 Pages with Accessibility-Ready Structure	Semantic HTML5, accessibility	CO1, CO2	PO1(3), PO2(2), PO3(3), PO4(2), PO5(3), PO9(1), PO10(3)	04
3	Responsive UI Design using CSS2.0/CSS3.0, CSS Grid and Flexbox	CSS Grid/Flexbox	CO2	PO1(3), PO2(2), PO3(3), PO4(2), PO5(3), PO9(1), PO10(3)	04
4	JavaScript, JavaScript DOM Manipulation, Event Handling, and UI Interactivity	DOM, events, localStorage	CO3, CO4	PO1(3), PO2(3), PO3(3), PO4(2), PO5(3), PO9(1), PO10(3)	04
5	Registration Form with Frontend Validation and User-Friendly Error Handling	Forms, JS validation	CO1, CO3	PO1(3), PO2(3), PO3(3), PO4(2), PO5(3), PO9(1), PO10(3)	04
6	Rendering External JSON Data using Fetch API, Search and Filter	JSON, Fetch API	CO3, CO4	PO1(3), PO2(3), PO3(3), PO4(2), PO5(3), PO9(1), PO10(3)	04
7	PHP Form Processing with Server-Side Validation and CSV/JSON File Storage	PHP form processing, CSV/JSON file	CO1, CO5	PO1(3), PO2(3), PO3(3), PO4(3), PO5(3), PO9(1), PO10(3), PO11(2), PO12(3)	04
8	MySQL Schema Design, ER Model and Database	MySQL schema	CO1,	PO1(3),	04

	Connectivity		CO5	PO2(3), PO3(3), PO4(3), PO5(3), PO9(1), PO10(3), PO11(2), PO12(3)	
9	Secure User Registration with Database Insert, Duplicate Email Check, and Password Hashing	Secure registration, hashing	CO1, CO3, CO5	PO1(3), PO2(3), PO3(3), PO4(3), PO5(3), PO9(1), PO10(3), PO11(2), PO12(3)	04
10	Secure Login/Logout with Sessions, Role-Based Access, Timeout, and Remember-Me Token	Sessions, RBAC, timeout	CO1, CO5	PO1(3), PO2(3), PO3(3), PO4(3), PO5(3), PO9(1), PO10(3), PO11(2), PO12(3)	02
11	Student Management CRUD with Search, Filter, and MySql Statements	Student CRUD	CO1, CO4, CO5	PO1(3), PO2(3), PO3(3), PO4(3), PO5(3), PO9(1), PO10(3), PO11(2), PO12(3)	04
12	Event Management CRUD with Poster Upload and File Validation	Event CRUD, file upload	CO1, CO4, CO5	PO1(3), PO2(3), PO3(3), PO4(3), PO5(3), PO9(1), PO10(3), PO11(2), PO12(3)	04
13	PHP JSON API and AJAX/Fetch-Based CRUD without Page Reload	PHP JSON API, AJAX/Fetch	CO3, CO4, CO5	PO1(3), PO2(3), PO3(3), PO4(3), PO5(3), PO9(1), PO10(3), PO11(2), PO12(3)	04
14	Admin Dashboard with Role-Based Access Control and Dynamic Menu	Admin dashboard	CO1, CO4, CO5	PO1(3), PO2(3), PO3(3), PO4(3), PO5(3), PO9(1), PO10(3), PO11(2),	06

				PO12(3)	
15	Final Integration, Testing, Documentation, GitHub Submission, and Local Deployment	Integration, testing, deployment	CO2, CO4, CO5	PO1(3), PO2(3), PO3(3), PO4(3), PO5(3), PO9(1), PO10(3), PO11(2), PO12(3)	04

2. Detailed Practical List

Practical 1: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition	Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.
Key Questions / Analysis / Interpretation	1. What is URL and parts of URL? 2. How does HTML file processed in web browser? 3. How will page navigation flow managed among all HTML pages? 4. How will GitHub commits be maintained after each practical?
Key Skills to be Addressed	Requirement analysis, sitemap design, wireframing, Git/GitHub, README writing, folder structuring
Applications	Web application planning and maintainable project setup
Learning Outcome	Students will be able to analyze requirements and prepare a structured project foundation before implementation.
CO Mapping	CO1
Dataset/Test Data	N/A; design artefacts only
Tools/Technology to be Used	VS Code, Draw.io/Figma, Git, GitHub
Total Hours	Implementation: 4 hours | Total engagement: 4 hours
Post Laboratory Work Description	Submit project functional requirements, sitemap, wireframe screenshots, GitHub repository link, and README.md.
Evaluation Strategy Including Viva	Requirement clarity, sitemap completeness, role definition, repository structure, and viva on project scope.
Advanced/Intermediate Extension	Intermediate: Create responsive wireframes in Figma. Advanced: Prepare ER diagram and REST-style route plan for future backend modules.

Practical 2: Semantic HTML5 Pages with Accessibility-Ready Structure

Problem Definition	Develop static HTML5 skeletons for at least 10 StudentHub pages such as Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ, and Feedback. Use semantic tags and accessibility-friendly structure.
Key Questions / Analysis / Interpretation	1. Are semantic tags such as header, nav, main, section, article, aside, and footer used properly? 2. Are form labels, headings, image alt text, and navigation links accessible? 3. Is the page structure consistent across all pages?
Key Skills to be Addressed	HTML5 semantic elements, accessibility basics, page linking, navigation menu, content hierarchy
Applications	Static website foundations and accessible UI design
Learning Outcome	Students will create clean, linked, and accessibility-ready HTML page structures.
CO Mapping	CO1, CO2
Dataset/Test Data	Sample page content for student portal
Tools/Technology to be Used	VS Code, HTML5, browser accessibility inspection tools
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit static page screenshots, source code, and accessibility checklist.
Evaluation Strategy Including Viva	Semantic correctness, page linking, accessibility readiness, and code readability.
Advanced/Intermediate Extension	Intermediate: Add breadcrumb navigation. Advanced: Add skip-to-content link and keyboard-friendly navigation.

Practical 3: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition	Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.
Key Questions / Analysis / Interpretation	1. How does the layout adapt for mobile, tablet, and desktop screens? 2. Which layout technique is used and why? 3. Are typography, spacing, colors, and contrast readable? 4. Is the framework used consistently without unnecessary inline CSS?
Key Skills to be Addressed	CSS3, media queries, Flexbox, Grid, Bootstrap components, responsive design
Applications	Multi-device web interface design
Learning Outcome	Students will design a professional, responsive, and reusable UI layout.
CO Mapping	CO2
Dataset/Test Data	N/A; UI design only
Tools/Technology to be Used	HTML5, CSS3, Bootstrap, browser responsive mode
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit screenshots for mobile, tablet, and desktop views.
Evaluation Strategy Including Viva	Responsiveness, framework usage, visual consistency, and CSS organization.
Advanced/Intermediate Extension	Intermediate: Add Contact and Feedback pages. Advanced: Create reusable layout components using JavaScript libraries such as jQuery.

Practical 4: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition	Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.
Key Questions / Analysis / Interpretation	1. How are DOM elements selected and modified? 2. Are event listeners attached correctly? 3. Is localStorage used for remembering theme choice? 4. Does interactivity improve usability without breaking accessibility?
Key Skills to be Addressed	DOM selection, event handling, class manipulation, localStorage, basic animation
Applications	Interactive dashboards and dynamic frontend behaviour
Learning Outcome	Students will apply JavaScript to enhance user experience and client-side interactivity.
CO Mapping	CO3, CO4
Dataset/Test Data	Static FAQ/event content
Tools/Technology to be Used	JavaScript ES6+, HTML5, CSS3, browser dev tools
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit live demo and source-code explanation.
Evaluation Strategy Including Viva	DOM logic, event handling correctness, usability, and browser console testing.
Advanced/Intermediate Extension	Intermediate: Add transition effects to cards/buttons. Advanced: Store user UI preferences in localStorage and restore them on page load.

Practical 5: Registration Form with Frontend Validation and User-Friendly Error Handling

Problem Definition	Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.
Key Questions / Analysis / Interpretation	1. Are correct input types and attributes used? 2. Are validation errors displayed near the relevant fields? 3. Is password strength checked? 4. Is the form accessible using keyboard and screen-reader-friendly labels?
Key Skills to be Addressed	HTML forms, JavaScript validation, regular expressions, password strength meter, UX error messages
Applications	Registration and data-entry systems
Learning Outcome	Students will design accessible forms with robust client-side validation.
CO Mapping	CO1, CO3
Dataset/Test Data	Valid and invalid sample registration inputs
Tools/Technology to be Used	HTML5, JavaScript ES5/ES6, browser dev tools
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit validation test cases with screenshots for valid and invalid inputs.
Evaluation Strategy Including Viva	Input coverage, validation accuracy, error handling, and code walkthrough.
Advanced/Intermediate Extension	Intermediate: Add real-time validation on keyup/change events. Advanced: Add custom CAPTCHA using Canvas.

Practical 6: Rendering External JSON Data using Fetch API, Search and Filter

Problem Definition	Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.
Key Questions / Analysis / Interpretation	1. How is JSON fetched, parsed, and rendered? 2. Which array methods are used for search, filter, sort, and pagination? 3. How are loading and error states handled? 4. How is modularity maintained in JavaScript files?
Key Skills to be Addressed	JSON, Fetch API, arrays, map/filter/sort, modular JavaScript, pagination logic
Applications	Dynamic data rendering in modern web applications
Learning Outcome	Students will consume external JSON data and render reusable frontend views.
CO Mapping	CO3, CO4
Dataset/Test Data	events.json, students.json, faqs.json with at least 15 records each
Tools/Technology to be Used	JavaScript ES6+, JSON, HTML5, CSS3
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit JSON files, rendered views, and test screenshots for search/filter/pagination.
Evaluation Strategy Including Viva	JSON parsing, array logic, UI rendering, error handling, and console testing.
Advanced/Intermediate Extension	Intermediate: Add dependent dropdowns for country/state/city. Advanced: Cache the latest JSON response in localStorage for offline-like display.

Practical 7: PHP Form Processing with Server-Side Validation and CSV/JSON File Storage

Problem Definition	Process submitted registration/contact form data using PHP. Validate and sanitize inputs on the server side and store records in CSV or JSON file format. Display success/error messages.
Key Questions / Analysis / Interpretation	1. Is the form submitted using POST? 2. Are inputs validated and sanitized server side? 3. Is file writing handled safely? 4. Are success and error responses displayed clearly?
Key Skills to be Addressed	PHP forms, POST handling, server-side validation, sanitization, file writing, CSV/JSON storage
Applications	Backend form processors and small data collection systems
Learning Outcome	Students will implement server-side form processing with safe file-based storage.
CO Mapping	CO1, CO5
Dataset/Test Data	Registration/contact form inputs
Tools/Technology to be Used	PHP, XAMPP/WAMP/LAMP, browser dev tools
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit PHP source code, test cases, and generated CSV/JSON file.
Evaluation Strategy Including Viva	POST handling, validation, sanitization, storage correctness, and viva.
Advanced/Intermediate Extension	Intermediate: Display stored CSV/JSON records on a webpage. Advanced: Add CSRF token validation for form submission.

Practical 8: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition	Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.
Key Questions / Analysis / Interpretation	1. Is the schema normalized properly? 2. Are primary keys and foreign keys correctly defined? 3. Is the database connection handled securely?
Key Skills to be Addressed	ER modeling, MySQL, phpMyAdmin, database error handling
Applications	Secure database-backed web applications
Learning Outcome	Students will design and connect a normalized database securely using PHP.
CO Mapping	CO1, CO5
Dataset/Test Data	SQL seed data for students, events, and registrations
Tools/Technology to be Used	MySQL, phpMyAdmin, PHP, XAMPP
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit SQL dump, db.php file, and connection test screenshot.
Evaluation Strategy Including Viva	Database, Database connections and DB viva.
Advanced/Intermediate Extension	Intermediate: Use PHP PDO and Prepared Statement. Advanced: Add stored procedural concept.

Practical 9: Secure User Registration with Database Insert, Duplicate Email Check, and Password Hashing

Problem Definition	Implement secure user registration by inserting user details into MySQL using MYSQLi. Validate input on frontend and backend, check duplicate email/username, hash passwords using password_hash(), and display appropriate messages.
Key Questions / Analysis / Interpretation	1. Is password_hash() used before storing passwords? 2. Is duplicate email/username checked? 3. Are frontend and backend validations consistent?
Key Skills to be Addressed	Secure registration, password hashing, validation, duplicate checks, insert operation
Applications	Secure account creation modules
Learning Outcome	Students will implement secure user registration following basic web security practices.
CO Mapping	CO1, CO3, CO5
Dataset/Test Data	Sample user registration inputs
Tools/Technology to be Used	PHP, MySQLi, XAMPP
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit registration test report with valid, invalid, and duplicate cases.
Evaluation Strategy Including Viva	Security implementation, DB insert correctness, test coverage, and viva.
Advanced/Intermediate Extension	Intermediate: Add prepared statement. Advanced: Add CSRF token and audit log entry for registration.

Practical 10: Secure Login/Logout with Sessions, Role-Based Access, Timeout, and Remember-Me Token

Problem Definition	Develop secure login and logout functionality using PHP sessions. Verify password using password_verify(), regenerate session ID after login, restrict dashboard access, implement role-based redirection, session timeout, and optional secure remember-me token.
Key Questions / Analysis / Interpretation	1. Are sessions started and destroyed securely? 2. Is password_verify() used correctly? 3. Is unauthorized dashboard access blocked? 4. Is session timeout implemented? 5. Are admin and student users redirected to appropriate dashboards?
Key Skills to be Addressed	PHP sessions, cookies, authentication, password verification, role-based access control, session timeout
Applications	Secure web authentication systems
Learning Outcome	Students will implement login/logout with secure session management.
CO Mapping	CO1, CO5
Dataset/Test Data	Seeded users table with hashed passwords and roles
Tools/Technology to be Used	PHP, MySQL, browser dev tools, XAMPP
Total Hours	Implementation: 2 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit screenshots for login success, failed login, logout, timeout, and unauthorized access attempt.
Evaluation Strategy Including Viva	Authentication correctness, session inspection, RBAC, timeout, and security viva.
Advanced/Intermediate Extension	Intermediate: Add last-login timestamp. Advanced: Implement remember-me using random token stored as hash in database; avoid custom JWT-like tokens.

Practical 11: Student Management CRUD with Search and Filter

Problem Definition	Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MYSQLi statements and clear success/failure messages.
Key Questions / Analysis / Interpretation	1. Are SELECT, INSERT, UPDATE, and DELETE operations working? 2. Is search by name/email/course implemented? 3. Are delete operations confirmed before execution?
Key Skills to be Addressed	PHP-MySQL CRUD, search, filter, UI feedback messages
Applications	Data-driven admin modules
Learning Outcome	Students will develop a complete and secure data-management page.
CO Mapping	CO1, CO4, CO5
Dataset/Test Data	students table with minimum 20 seeded records
Tools/Technology to be Used	PHP, MySQL, XAMPP
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit SQL dump, CRUD screenshots, and test cases.
Evaluation Strategy Including Viva	CRUD correctness, query safety, UI linkage, pagination/search logic, and viva.
Advanced/Intermediate Extension	Intermediate: Export student list to CSV. Advanced: Add soft delete using status column instead of permanent delete.

Practical 12: Event Management CRUD with Poster Upload and File Validation

Problem Definition	Develop event management functionality for admin users. Admin can add, view, update, delete, and upload event posters with file type and file size validation.
Key Questions / Analysis / Interpretation	1. Are add, update, and delete functionalities implemented correctly? 2. Is the UI linked with the database correctly? 3. Are poster uploads validated for type and size? 4. Are success and failure messages displayed?
Key Skills to be Addressed	PHP-MySQL CRUD, file upload, server-side validation, event status management, admin UI
Applications	Event and content management systems
Learning Outcome	Students will implement a secure CRUD module with validated file upload.
CO Mapping	CO1, CO4, CO5
Dataset/Test Data	events table with sample event records and optional poster images
Tools/Technology to be Used	PHP, MySQL, Bootstrap/Tailwind, XAMPP
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit CRUD demo screenshots, upload validation screenshots, and SQL dump.
Evaluation Strategy Including Viva	CRUD logic, file validation, UI feedback, access restriction, and code review.
Advanced/Intermediate Extension	Intermediate: Add event category and open/closed filter. Advanced: Add image renaming, upload directory protection, and audit log for event changes.

Practical 13: PHP JSON API and AJAX/Fetch-Based CRUD without Page Reload

Problem Definition	Create simple PHP JSON API endpoints for events or students and consume them using Fetch API. Perform list, add, update, and delete operations asynchronously without full page reload.
Key Questions / Analysis / Interpretation	1. Do PHP endpoints return valid JSON responses? 2. Are GET and POST requests handled properly? 3. Are loading, success, and error states shown on the frontend? 4. Is input validation performed before database operations?
Key Skills to be Addressed	PHP JSON API, Fetch API, asynchronous CRUD, JSON response handling, frontend-backend integration
Applications	Modern interactive web applications and API-driven dashboards
Learning Outcome	Students will integrate frontend JavaScript with PHP backend using JSON APIs.
CO Mapping	CO3, CO4, CO5
Dataset/Test Data	students/events API connected to MySQL tables
Tools/Technology to be Used	PHP, MySQL, JavaScript Fetch API, browser network tab
Total Hours	Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit API endpoint files, browser network screenshots, and AJAX CRUD demo.
Evaluation Strategy Including Viva	API response format, asynchronous behaviour, DB correctness, error handling, and viva.
Advanced/Intermediate Extension	Intermediate: Add AJAX search with debounce. Advanced: Add API response status codes and centralized error handling.

Practical 14: Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log

Problem Definition	Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.
Key Questions / Analysis / Interpretation	1. Are dashboard values loaded dynamically from the database? 2. Is access restricted to admin users? 3. Are role-based menus working? 4. Are important admin actions recorded in audit_logs?
Key Skills to be Addressed	Dashboard design, RBAC, SQL aggregation, audit logging, dynamic menu loading
Applications	Administrative monitoring and moderation panels
Learning Outcome	Students will build a role-based admin dashboard with basic monitoring and traceability.
CO Mapping	CO1, CO4, CO5
Dataset/Test Data	users, students, events, registrations, audit_logs tables
Tools/Technology to be Used	PHP, MySQL, Bootstrap, JavaScript charts optional
Total Hours	Implementation: 6 hours | Total engagement: 6 hours
Post Laboratory Work Description	Submit admin dashboard screenshots and audit-log test cases.
Evaluation Strategy Including Viva	RBAC correctness, dashboard queries, audit log entries, UI quality, and viva.
Advanced/Intermediate Extension	Intermediate: Add active/inactive user status toggle. Advanced: Add simple analytics chart for event registrations by month.

Practical 15: Final Integration, Testing, Documentation, GitHub Submission, and Local Deployment

Problem Definition	Integrate all modules into a single deployable StudentHub portal. Verify navigation, responsiveness, authentication, database operations, security checks, AJAX functionality, and documentation. Deploy locally using XAMPP/Apache virtual host and submit GitHub repository.
Key Questions / Analysis / Interpretation	1. Are all pages properly linked and navigable? 2. Are sessions, roles, validations, and database operations working end-to-end? 3. Are security and accessibility checklists completed? 4. Is project documentation sufficient for installation and testing? 5. Is the project repository clean and properly committed?
Key Skills to be Addressed	Full-stack integration, testing, deployment, documentation, GitHub, debugging
Applications	Complete deployable web application project
Learning Outcome	Students will deliver a complete, secure, documented, and testable full-stack web application.
CO Mapping	CO2, CO4, CO5
Dataset/Test Data	Complete semester project database and test inputs
Tools/Technology to be Used	All technologies used: HTML, CSS, Bootstrap/Tailwind, JavaScript, PHP, MySQL, GitHub, XAMPP
Total Hours	Implementation: 4 hours | Total engagement: 4 hours
Post Laboratory Work Description	Submit final GitHub link, SQL dump, README, test report, screenshots, and project demo.
Evaluation Strategy Including Viva	Holistic viva, integrated demo, security/accessibility checklist, documentation quality, and deployment test.
Advanced/Intermediate Extension	Intermediate: Deploy locally with Apache Virtual Hosts. Advanced: Deploy frontend/static part online and document backend deployment limitations or use a PHP-compatible hosting environment.

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